

Exercise on Logic Circuit Design

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1 Conversion

Question. Convert from decimal to 2-complement 8-bit binary numbers.

- (a) 105 (b) -28 (c) 76
(d) -25 (e) -101 (f) 127

2 Subtraction of 2-complement binary numbers

Question. Subtract the following 8-bit, 2-complement binary numbers. State if there is an overflow or an underflow.

- (a)
$$\begin{array}{r} 1\ 0\ 0\ 1\ 0\ 0\ 1\ 1 \\ -\ 0\ 1\ 1\ 0\ 1\ 1\ 0\ 1 \\ \hline \end{array}$$
 (b)
$$\begin{array}{r} 1\ 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ -\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$
 (c)
$$\begin{array}{r} 0\ 1\ 0\ 1\ 1\ 0\ 1\ 1 \\ -\ 0\ 1\ 0\ 1\ 1\ 0\ 1\ 0 \\ \hline \end{array}$$

(d)
$$\begin{array}{r} 1\ 0\ 0\ 1\ 1\ 0\ 1\ 1 \\ -\ 1\ 1\ 0\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$
 (e)
$$\begin{array}{r} 1\ 1\ 1\ 1\ 1\ 1\ 0\ 1 \\ -\ 0\ 1\ 1\ 0\ 1\ 1\ 0\ 1 \\ \hline \end{array}$$
 (f)
$$\begin{array}{r} 1\ 1\ 0\ 1\ 0\ 0\ 1\ 1 \\ -\ 1\ 1\ 1\ 0\ 0\ 0\ 1\ 1 \\ \hline \end{array}$$

3 Subtraction of BCD numbers

Question.

- (a)
$$\begin{array}{r} 0\ 1\ 1\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0\ 0\ 0 \\ -\ 0\ 0\ 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 1 \\ \hline \end{array}$$

(b)
$$\begin{array}{r} 0\ 0\ 1\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 1\ 0\ 1\ 1\ 1 \\ -\ 0\ 0\ 0\ 1\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 0\ 0 \\ \hline \end{array}$$

(c)
$$\begin{array}{r} 1\ 0\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 1\ 1 \\ -\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 1\ 0 \\ \hline \end{array}$$

(d)
$$\begin{array}{r} 0\ 1\ 1\ 0\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 1\ 0\ 1\ 0\ 1 \\ -\ 0\ 1\ 1\ 0\ 1\ 0\ 0\ 0\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 1 \\ \hline \end{array}$$